Graph machine learning for flight delay prediction due to holding maneuver

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Abstract

Flight delays caused by holding maneuvers represent a critical and costly challenge in modern aviation systems. These delays typically arise due to air traffic congestion, adverse weather conditions, and air traffic control (ATC) restrictions, requiring aircraft to circle within designated airspace before landing. While such maneuvers are essential for maintaining safety, they significantly increase fuel consumption, environmental emissions, and passenger dissatisfaction, making accurate delay prediction crucial for improving operational efficiency.

In this work, the problem of predicting flight delays due to holding maneuvers is modeled as a graph-based learning task to effectively capture the complex spatial and temporal interdependencies present in air traffic networks. Unlike traditional machine learning approaches that rely on tabular data and often fail to represent relational structures, this study leverages Graph Machine Learning (Graph ML) techniques. Specifically, we implement two approaches: (1) CatBoost enhanced with graph-derived features such as network centrality and connectivity, and (2) Graph Attention Networks (GATs), which directly learn from graph-structured data.

Experimental results demonstrate that the CatBoost model outperforms GATs in handling imbalanced datasets, providing more accurate predictions of holding events. Additionally, CatBoost offers improved interpretability through feature importance analysis, enabling better understanding of contributing factors. Furthermore, the study explores the practical applicability of the model through a web-based tool for real-time delay prediction and simulation.

Overall, the findings highlight the effectiveness of graph-based approaches in aviation analytics, with significant implications for enhancing fuel efficiency, reducing delays, and improving passenger experience.

1. Introduction

The aviation industry increasingly relies on data-driven methodologies to enhance operational efficiency and minimize delays. One of the major challenges in air traffic management is the occurrence of holding maneuvers, where aircraft are required to delay landing by circling within designated airspace. These maneuvers are typically caused by airport congestion, adverse weather conditions, and air traffic control (ATC) constraints. Although essential for maintaining safety, holding patterns contribute significantly to increased fuel consumption, environmental emissions, and passenger dissatisfaction, making accurate prediction of such delays a critical task in modern aviation systems.

Traditional approaches to flight delay prediction have primarily focused on machine learning models applied to tabular datasets, incorporating features such as weather conditions, airport congestion levels, and flight schedules [1], [2]. While these methods have demonstrated reasonable performance, they often fail to capture the complex spatial and temporal interdependencies inherent in air traffic networks. Recent studies have identified class imbalance as a persistent issue in delay prediction tasks, further limiting model performance [3]. To address these limitations, network-based approaches have emerged as a promising alternative, enabling the modeling of relationships between airports, flights, and airspace constraints [4].

Graph Machine Learning (Graph ML) and Graph Neural Networks (GNNs) have gained significant attention for their ability to represent and learn from relational data structures. In the context of aviation, these techniques allow for a more comprehensive understanding of interconnected systems, where delays at one node (airport or flight) can propagate across the network [5], [6]. The relationship between departure delays, en-route conflicts, and holding maneuvers is particularly important, as holding is often a direct consequence of congestion and conflict resolution in controlled airspace [7].

Recent advancements in intelligent transportation systems have demonstrated the effectiveness of graph-based models in capturing both spatial and temporal dependencies. For instance, Rahmani et al. [8] highlight the broad applicability of GNNs in domains such as traffic forecasting, demand prediction, and urban planning, emphasizing their ability to model highly interconnected systems. Similarly, Zhao et al. [9] proposed a Temporal Graph Convolutional Network (T-GCN) that combines Graph Convolutional Networks (GCNs) with Gated Recurrent Units (GRUs), achieving state-of-the-art performance in real-time traffic prediction tasks. These studies underscore the potential of graph-based techniques in addressing complex, dynamic problems in transportation systems.

Motivated by these advancements, this study models the prediction of flight delays due to holding maneuvers as a graph-based learning problem. Specifically, we explore two complementary approaches:

1. **Tabular-based approach:** A CatBoost model enhanced with graph-derived features such as centrality and connectivity metrics, which capture the importance of nodes and edges within the air traffic network [10].
2. **Graph-based approach:** A Graph Attention Network (GAT) model that directly learns from graph-structured data, leveraging attention mechanisms to capture the relative importance of neighboring nodes [11].

The primary contributions of this work are threefold. First, we demonstrate the effectiveness of graph-based feature engineering in improving the accuracy of holding delay prediction. Second, we provide a comparative analysis between CatBoost and GAT models, highlighting their strengths in handling imbalanced datasets and relational data. Third, we discuss the practical implications of the proposed models, including their potential integration into real-time decision-support systems for air traffic management.

The remainder of this report is organized as follows. Section 2 reviews the relevant literature on flight delay prediction and graph-based learning approaches. Section 3 describes the dataset and methodology used in this study. Section 4 presents the experimental results and performance evaluation. Finally, Section 5 concludes the report with key findings and future research directions.

2. Theoretical Framework and Related Work

Graph Machine Learning (Graph ML) has emerged as a powerful paradigm for learning from graph-structured data, enabling tasks such as node classification, edge prediction, and network embedding [12]–[14]. Unlike traditional machine learning methods that operate on independent data points, Graph ML captures the relationships and dependencies between entities, making it particularly suitable for complex systems such as air traffic networks. Existing approaches in this domain can be broadly categorized into traditional graph-based feature engineering methods and deep graph learning techniques based on Graph Neural Networks (GNNs) [15]–[17].

2.1 Traditional Graph Learning

Traditional graph learning approaches focus on extracting structural features from graphs and integrating them into standard machine learning models [22], [23]. These features include centrality measures such as degree centrality, betweenness centrality, and eigenvector centrality, which quantify the importance of nodes within a network [24]. Extensions of these concepts to edges have also been explored to better represent relationships in directed graphs [25], [26].

One widely used algorithm in this context is PageRank, which extends eigenvector centrality by introducing a damping factor to model random transitions within the network [27]. It is mathematically defined as:

$$PR(v) = \frac{1-d}{N} + d \sum_{u \in \mathcal{N}(v)} \frac{PR(u)}{\deg(u)},$$

where (d) is the damping factor, (N(v)) represents the set of neighboring nodes, and (\deg(u)) denotes the degree of node (u).

Despite their effectiveness, traditional graph-based features have limitations. Spectral centrality measures often exhibit high computational complexity, typically on the order of (O(n^2d)), and primarily focus on node-level representations. As a result, they may fail to adequately capture edge-level dynamics and directional relationships, which are critical in modeling air traffic systems.

2.2 Deep Graph Learning

Deep learning approaches on graphs, particularly Graph Neural Networks (GNNs), have significantly advanced the field by enabling end-to-end learning from graph-structured data. GNNs are generally categorized into two types: spectral-based methods and spatial-based methods [19], [20].

2.2.1 Spectral-based GNNs

Spectral-based methods rely on graph signal processing techniques, where convolution operations are defined in the spectral domain using the graph Laplacian (L = D - A), where (D) is the degree matrix and (A) is the adjacency matrix [28]. Graph Convolutional Networks (GCNs) simplify earlier spectral approaches by approximating localized filters, leading to efficient formulations such as:

2.2.2 Spatial-based GNNs

Spatial-based GNNs operate directly on graph structures by performing neighborhood aggregation through message-passing mechanisms. These models learn node representations by iteratively combining information from neighboring nodes, making them more flexible and applicable to real-world graphs.

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{Wh}_i \parallel \mathbf{Wh}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{Wh}_i \parallel \mathbf{Wh}_k]))},$$

where (W) is a learnable weight matrix, (a) is the attention vector, and (hi), (hj) represent node features. This formulation enables adaptive weighting of relationships, allowing the model to capture heterogeneous dependencies within complex networks.

3. Materials and Methods

This section presents the datasets, feature engineering strategies, and modeling approaches employed in this study. The primary objective is to predict whether a flight will experience delays due to holding maneuvers by leveraging both traditional machine learning and graph-based deep learning techniques. Specifically, we compare a CatBoost model enhanced with graph-derived features and a Graph Attention Network (GAT) model that directly operates on graph-structured data.

3.1 Dataset Description

The study utilizes two datasets comprising 42,336 flight operations within the Brazilian airspace system, covering the period from June 1, 2022, to February 16, 2023 (260 operational days) [29]. Each observation corresponds to a single flight and includes meteorological, geographical, and operational attributes.

For the classification task, the dataset labels flights based on the occurrence of holding maneuvers, resulting in 720 positive samples (holding events) and 41,616 negative samples (non-holding events), yielding a significant class imbalance (~1.7%). Additionally, a regression dataset is constructed to predict the duration of holding (in seconds) for delayed flights.

The dataset incorporates three main categories of features:

- **Meteorological features:** Derived from METAR and METAF reports, including wind speed and direction, visibility, temperature, and cloud coverage [1].
- **Geographical features:** Include latitude, longitude, airport altitude, and geodesic distance between origin and destination airports.
- **Operational features:** Capture flight-specific attributes such as scheduled departure time, runway usage, and air traffic conditions.

These features collectively provide a comprehensive representation of the operational environment influencing holding maneuvers.

3.2 Data Preprocessing

The dataset contains minimal missing values. For meteorological variables, missing entries are handled using forward-filling based on the most recent valid observations, aligning with standard aviation data practices. Graph-based features are computed from a complete network representation, ensuring no missing values in structural attributes.

To ensure robust evaluation, a two-stage data splitting strategy is adopted. First, the dataset is divided into training (80%) and test (20%) sets. The training set is further split into training (70%) and validation (10%) subsets using a fixed random seed to ensure reproducibility. This results in a final 70–10–20 split for training, validation, and testing, respectively. Additionally, temporal validation experiments are conducted to assess the model's generalization capability in real-world deployment scenarios.

3.3 Graph Representation of Flight Network

In this **fig. 1** , each directed edge corresponds to a flight, capturing both connectivity and directionality. Since multiple flights may occur between the same pair of airports, the system is initially modeled as a multigraph. To facilitate feature extraction for traditional ML models, the multigraph is transformed into a weighted directed graph, where edge weights represent the number of flights between airport pairs.

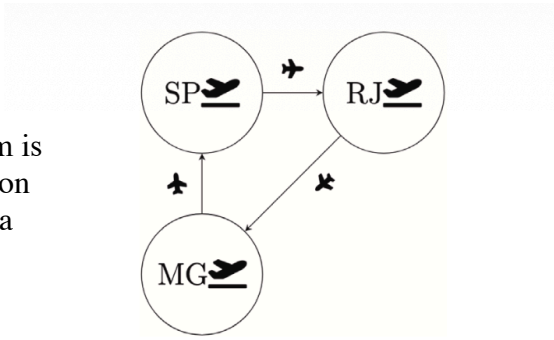

Fig. 1. Example of airports and directed flights.

3.4 CatBoost with Graph-Based Features

The first modeling approach employs CatBoost, a gradient boosting decision tree algorithm known for its robustness in handling categorical features, noisy data, and class imbalance [10]. CatBoost is particularly suitable for structured datasets and provides interpretability through feature importance analysis.

Graph-based features are extracted from the weighted directed graph to enhance predictive performance. These include:

1. **Betweenness Centrality:** Measures the importance of an airport in terms of shortest paths within the network.
2. **Flow Betweenness:** Captures traffic flow dynamics across the network [31].
3. **Edge Connectivity:** Reflects the robustness of connections between airports.
4. **Degree Difference:** Indicates disparities between incoming and outgoing traffic.
5. **Google Matrix (PageRank-based):** Provides probabilistic transition insights into network connectivity [27].

To compute transition probabilities, we define:

$$p_{ij} = \frac{a_{ij}}{\sum_k a_{ik}}$$

representing the probability of transitioning from node i to node j in a random walk on the flight network. The Google Matrix G is then defined as:

$$G = \alpha P + (1-\alpha)\frac{ee^T}{n}$$

3.5 Graph Attention Network (GAT) Model

The second approach utilizes Graph Attention Networks (GATs), which directly operate on graph structured data and learn node representations through attention-based message passing **fig. 2** .

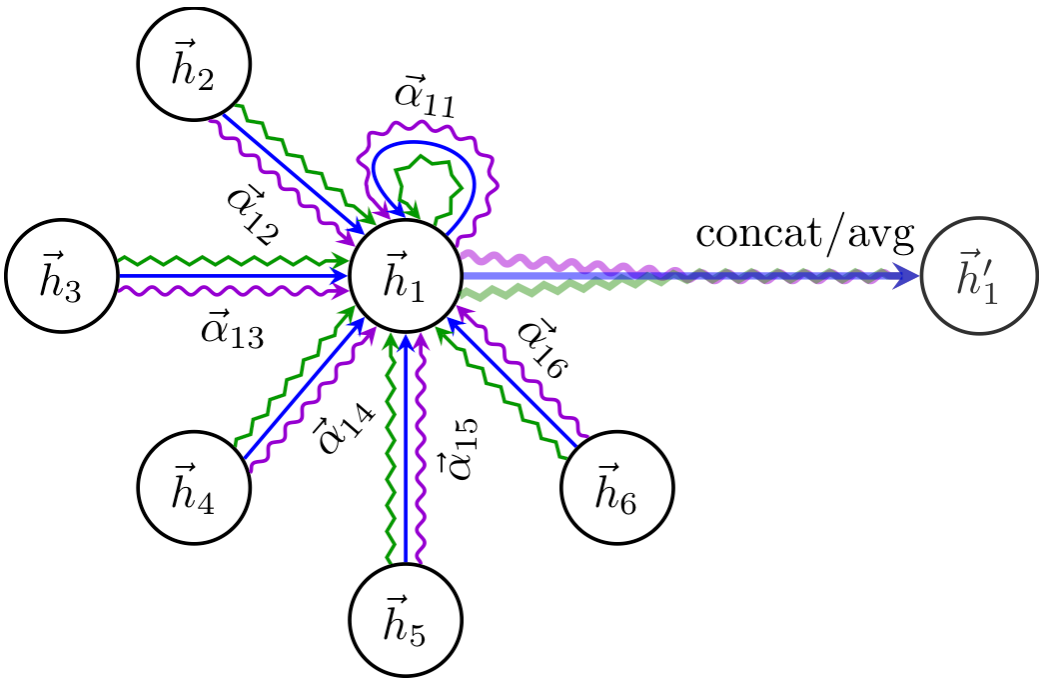

Fig. 2. Airport multigraph GAT Layer with multi-head attention for three different flights between nodes (SP,RJ), with alternating colors opacity for each flight.

Unlike CatBoost, GAT does not rely on explicitly engineered graph features. Instead, it learns representations end-to-end by assigning attention weights to neighboring nodes:

$$\alpha_{ij} = \sigma(\phi_1(\mathbf{a}^T[Wh_i \| Wh_j \| W_2 e_{ij}])),$$

$$\alpha_{ij}^{(k)} = \frac{\exp\left(\sigma\left(\mathbf{a}^T[Wh_i \| Wh_j \| W_2 e_{ij}^{(k)}]\right)\right)}{\sum_{l\in\mathcal{N}_i}\sum_{m=1}^{K_{il}} \exp\left(\sigma\left(\mathbf{a}^T[Wh_i \| Wh_l \| W_2 e_{il}^{(m)}]\right)\right)}$$

To handle multiple flights between the same airport pairs, a multigraph attention mechanism is employed, allowing each edge to contribute independently. The node embeddings are updated as:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}_i}\sum_{k=1}^{K_{ij}} \alpha_{ij}^{(k)} \cdot Wh_j^{(l)}\right)$$

The final prediction is obtained by combining node embeddings with edge features through a multilayer perceptron (MLP).

The GAT model is trained using:

- Hidden dimension: 32
- Configurable layers (1–30)
- Adam optimizer
- Learning rate: 0.01
- Focal loss ($\alpha = 0.1$, $\gamma = 7.5$) to address class imbalance

3.6 Model Evaluation

Model performance is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. Experimental results indicate that the CatBoost model achieves higher stability and better handling of class imbalance compared to GAT models, particularly in deeper architectures **fig. 3**.

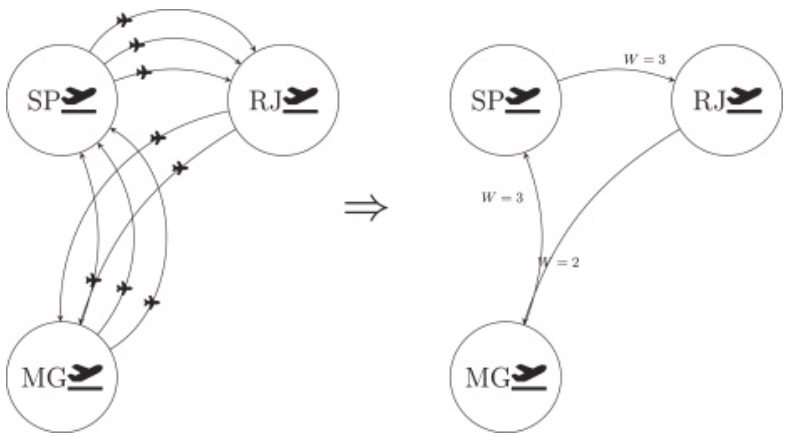

Fig. 3. Transformation of a multigraph of flights into a weighted directed graph. The multigraph (left) represents multiple flights between airports. In the weighted graph (right), edges are aggregated to show total flights as weights.

Table 1
Performance metrics for various GAT layer configurations and CatBoost with graph features.

Model	Test accuracy	Precision	Recall	F1-Score
CatBoost	0.90	**0.09**	0.58	**0.16**
1 GAT Layer	**0.95**	0.03	0.06	0.04
3 GAT Layers	0.52	0.01	0.40	0.03
5 GAT Layers	0.57	0.01	0.30	0.02
10 GAT Layers	0.91	0.02	0.08	0.03
30 GAT Layers	0.02	0.02	**0.99**	0.03

4. Results

This section presents a comprehensive evaluation of the proposed models, focusing on the comparative performance of the CatBoost and Graph Attention Network (GAT) approaches. Additionally, we analyze the regression capabilities and interpretability of the CatBoost model using graph-derived features.

4.1 Model Performance and Interpretability

The performance of the models in predicting flight delays due to holding maneuvers was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. As summarized in Table 1, the single-layer GAT model achieved the highest accuracy; however, it exhibited very low precision and F1-score. This indicates that, despite correctly classifying a large number of non-delay instances, the model struggled to identify true positive holding events, which is critical in highly imbalanced datasets.

In contrast, the CatBoost model demonstrated a more balanced performance across all evaluation metrics, making it more suitable for this application. Its ability to better capture minority-class instances (holding events) highlights its robustness in handling class imbalance, a known challenge in aviation delay prediction tasks [3], [10].

As illustrated in **Fig. 4,** feature importance analysis reveals that operational and meteorological features—such as METAF altimeter readings, wind direction, wind speed, and time of flight—are the most influential predictors. Graph-based features, including betweenness centrality and Google matrix scores, also contribute meaningfully, though to a lesser extent. This demonstrates that while network topology provides valuable contextual information, flight-specific and environmental factors remain dominant drivers of holding delays.

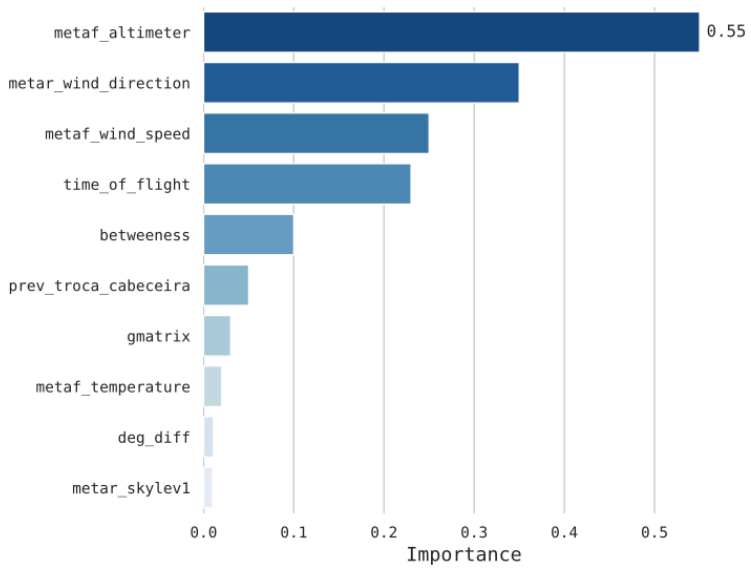

Fig. 4. Feature importance for the CatBoost model on the airport network dataset, highlighting the relevance of graph-based features.

Beyond classification, the CatBoost model was also evaluated on a regression task aimed at predicting continuous delay durations. The distributions of predicted and actual delay values are shown in **Fig. 5** and **6**, respectively.

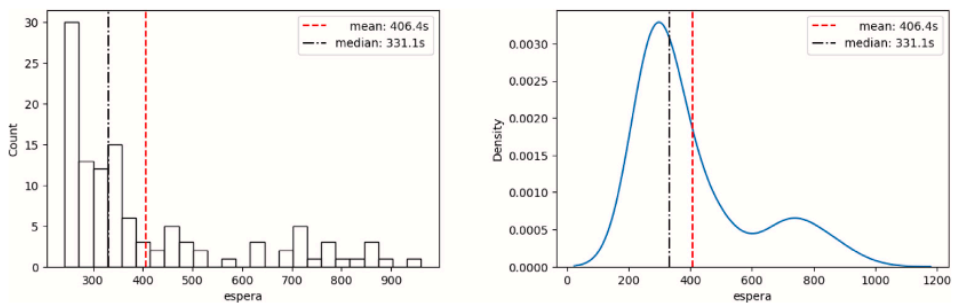

Fig. 5. Distribution of predicted delay values (y_{pred}) for the regression task using CatBoost.

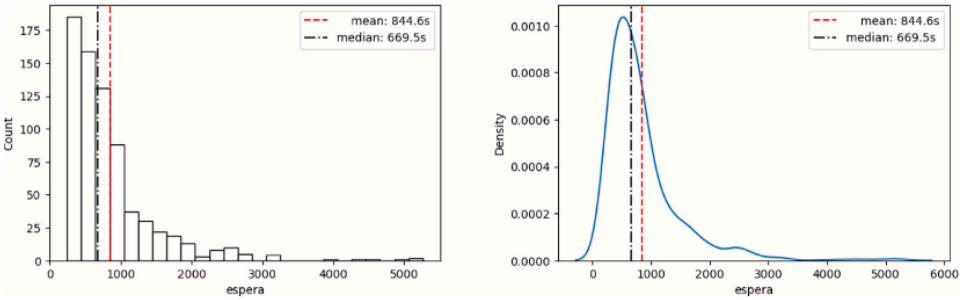

Fig. 6. Distribution of actual delay values (y_{test}) for the test set in the regression task.

Figures 5 and 6 further illustrate the alignment between predicted and actual delay distributions. The predicted delays exhibit a similar shape to the ground truth distribution, indicating that the model successfully captures the underlying trends in delay patterns. However, discrepancies at extreme values suggest that the model may underestimate rare but severe delay events. This limitation highlights the need for further refinement using metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and R^2, which can provide deeper insights into regression performance [9].

Interpretability is a key advantage of the CatBoost approach. By leveraging graph-based features alongside traditional variables, the model provides transparency through feature importance rankings. Explainable AI (XAI) techniques help identify how different features contribute to predictions, enabling better decision-making in operational environments. Notably, while graph metrics appear among the top predictors, operational variables such as flight timing and weather conditions dominate, suggesting that network structure complements rather than replaces traditional features.

4.2 Deployment, Implementation, and Discussion

The practical applicability of the proposed approach is demonstrated through the development of a web-based simulation tool, *Airdelay*, which enables real-time visualization and prediction of flight delays. The system is implemented using modern web technologies and integrates the trained CatBoost model to simulate various operational scenarios.

The tool allows users to:

- Visualize airport networks and flight connections
- Simulate holding delays under different weather and traffic conditions
- Analyze model predictions interactively

This deployment highlights the real-world usability of the proposed framework in decision-support systems for air traffic management.

Overall, the results emphasize the superiority of the CatBoost model over GAT in this context. While GAT models show potential in capturing relational dependencies, they are more sensitive to class imbalance and prone to overfitting in deeper architectures. These findings align with prior research indicating that ensemble-based methods often outperform deep learning models in structured and imbalanced datasets [10], [11].

However, the study has certain limitations. The scope of hyperparameter tuning for GAT models was limited, and class imbalance remains a significant challenge. Future work should explore advanced techniques such as cost-sensitive learning, data augmentation, and alternative architectures, including multigraph neural networks [33], to further enhance performance.

5. Conclusion

This study investigated the prediction of flight delays caused by holding maneuvers using graph-based machine learning techniques. By modeling the air traffic system as a directed network, we explored two complementary approaches: a CatBoost model enhanced with graph-derived features and a Graph Attention Network (GAT) capable of learning directly from graph-structured data.

The experimental results demonstrate that while GAT models provide a flexible and expressive framework for capturing relational dependencies in complex networks, their performance is limited in highly imbalanced datasets. In contrast, the CatBoost model consistently achieved superior and more balanced performance across evaluation metrics, highlighting its effectiveness in handling class imbalance and structured tabular data [10], [11]. The integration of graph-based features further improved predictive capability, confirming the importance of incorporating network topology into aviation analytics.

In addition to classification, the regression analysis showed that the CatBoost model effectively captures the overall distribution and trends of delay durations, although some limitations remain in accurately predicting extreme delay values. These findings emphasize the need for further refinement through advanced regression techniques and robust evaluation metrics such as MAE, RMSE, and R^2 [9].

A key contribution of this work lies in its practical applicability. The development of the *Airdelay* web-based simulation tool demonstrates how predictive models can be integrated into real-time decision-support systems for air traffic management. This tool enables dynamic visualization and scenario analysis, providing valuable insights for operational planning and delay mitigation.

Despite these contributions, several limitations remain. The presence of significant class imbalance poses challenges for both traditional and deep learning models. Future work should explore advanced data balancing techniques such as GraphSMOTE [34], as well as alternative architectures tailored for multigraph and edge-level prediction tasks [33]. Additionally, incorporating topological learning methods, including persistent homology and topology-aware message passing, may further enhance the model's ability to capture complex network dynamics [35]–[37].

Furthermore, hybrid approaches that combine the strengths of gradient boosting and graph neural networks present a promising direction for improving predictive performance. The integration of real-time air traffic and weather data streams could also enhance the robustness and deployment readiness of such systems. From an operational perspective, adherence to validation frameworks and human-in-the-loop decision-making processes remains essential for ensuring reliability and safety in aviation applications [38], [39].

In conclusion, this study demonstrates that graph-based machine learning provides a powerful framework for modeling and predicting flight delays due to holding maneuvers. The results highlight the effectiveness of combining graph-theoretic insights with robust machine learning techniques, offering a pathway toward more efficient, reliable, and intelligent air traffic management systems.

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